

ROLLER_CONVEYOR_DESIGN_RULES.md

<!-- Machine-readable reference document. Intended to be loaded as context/system knowledge by an AI model generating or validating parametric roller-conveyor CAD via the Fusion API. Format: explicit constants, unambiguous rules, formulas. Avoid re-deriving anything marked FIXED. Every rule has an ID for citation in generated code comments or error messages. -->

0. PURPOSE

This document is the single source of truth for roller conveyor geometry generation.

Any generated model, script, or parameter set MUST conform to every rule below.

If a rule and a user input conflict, flag the conflict – do not silently override a FIXED value.

1. TERMINOLOGY (per CEMA 102 – use exactly these terms in code, comments, UI)

Term	Definition
BF	Between Frames – clear width between inner faces of the two side rails
OAW	Overall Width = $BF + 2 \times \text{railThickness} + 2 \times \text{railFlangeProjection}$

Term	Definition
RollFace / RollLength	Load-bearing tube length of the roller (excludes pin)
Pitch (P)	Center-to-center spacing between adjacent roller axes
BFH	Between-Frame Height — top of rail to ground

2. PARAMETER CLASSES

Every parameter in this system belongs to exactly one class. A parameter must never move between classes at generation time.

- **FIXED** — set once from the table below; identical across all instances of that part in one model
- **VARIABLE** — supplied by user per generation run; may differ between runs
- **DERIVED** — computed from FIXED + VARIABLE only, via the formulas in Section 5

RULE 2.1: A FIXED parameter must never appear as the output of a formula that includes a VARIABLE.

RULE 2.2: A DERIVED parameter must never be hand-edited after computation; regenerate instead.

3. FIXED PARAMETERS (source: this project's standard, based on CEMA 401

practice)

```
railThickness_mm: 3.0          # FIXED – single
value, both rails, constant along full length
railProfileHeight_mm: 40.0     # FIXED
legProfile: "40x40"           # FIXED – catalog
cross-section, not custom sketch
legMaterial: "Aluminum 6063-T5" # FIXED – per EN
755-2 / ASTM B221
rollerBodyDia_mm: 50.0        # FIXED – choose
one from {35, 48, 50, 63, 89}
rollerTubeGauge_mm: 1.5       # FIXED
rollerPinDia_mm: 8.0          # FIXED
rollerPinLength_mm: 12.0      # FIXED – length of
pin protruding beyond RollLength
pinEngagementClearance_mm: 0.3 # FIXED – total
clearance, pin OD to rail hole ID
rollerPitch_mm: 75.0          # FIXED DEFAULT –
user may override, see RULE 5.3
toleranceClass: "ISO 2768-mK" # FIXED – general
tolerance, all non-mating dimensions
```

RULE 3.1: Do not change values in this table per-instance. If a different duty class is needed (light/medium/heavy), define a second complete table – never blend values from two classes.

4. VARIABLE PARAMETERS (user-supplied per run)

```
conveyorLength_mm: <user input>
conveyorWidth_BF_mm: <user input>    # this is BF,
not OAW – see Section 1
legHeight_mm: <user input>
```

```
rollerPitch_override_mm: <optional>    # if present,
overrides FIXED rollerPitch_mm for this run only
```

5. DERIVED FORMULAS (compute only from FIXED + VARIABLE; never hardcode)

```
RULE 5.1  OAW_mm = conveyorWidth_BF_mm +
2*railThickness_mm + 2*railFlangeProjection_mm

RULE 5.2  rollerBodyLength_mm = conveyorWidth_BF_mm
- 2*pinEngagementClearance_mm
        # This is the CAD rule for "inner face of
rail touches roller pin base":
        # rail inner face position = ±
(rollerBodyLength_mm / 2) from centerline

RULE 5.3  effectivePitch_mm =
rollerPitch_override_mm OR rollerPitch_mm (FIXED
default)

RULE 5.4  rollerCount = floor((conveyorLength_mm -
endClearance_mm) / effectivePitch_mm) + 1

RULE 5.5  MINIMUM LOAD-SUPPORT CHECK (hard
constraint, not advisory):
        shortest expected unit load (mm) must span
>= 3 * effectivePitch_mm
        If violated: reduce effectivePitch_mm and
recompute RULE 5.4. Do not proceed silently.
```

6. GEOMETRY / ASSEMBLY RULES (hard constraints)

RULE 6.1 – Side rails: single extrude, one cross-section sketch, swept the full `conveyorLength_mm` with zero taper or scale change. `railThickness_mm` is read from Section 3 exactly once.

RULE 6.2 – Left and right rails are mirror instances of the same body/parameter set. Never two independently-dimensioned sketches.

RULE 6.3 – Legs: extrude profile = `legProfile` (Section 3), constant cross-section along `legHeight_mm` only. All legs in one model reference the same profile instance.

RULE 6.4 – Roller-to-rail joint: rigid/pin joint, zero gap, at the position computed in RULE 5.2.
Do not position by manual coordinate offset – use a Fusion joint referencing the named parameters.

RULE 6.5 – Leg-to-rail joint: rigid joint, at a FIXED offset from the rail profile (not recomputed from `railThickness_mm` per instance).

RULE 6.6 – No component may use a non-uniform Scale feature to force a dimensional fit. A fit failure means a parameter value is wrong (see Section 3/5) – fix the parameter, regenerate.

RULE 6.7 – Tolerancing: apply `toleranceClass` (Section 3) to all dimensions by default. Apply a tighter explicit fit only at: (a) roller pin OD ↔ rail hole ID, (b) leg-to-rail mounting face.

7. VALIDATION CHECKLIST (run before returning any generated model)

```
[ ] railThickness_mm referenced from Section 3
    exactly once, used by both rails
[ ] legProfile identical across every leg instance
    in the model
[ ] rail inner face coincides with roller pin base
    per RULE 5.2 (joint-verified, zero gap)
[ ] no part uses non-uniform scale
[ ] no FIXED value was recomputed inside a loop over
    length/count
[ ] RULE 5.5 load-support check passed for the given
    conveyorLength_mm / effectivePitch_mm
[ ] BF vs OAW used correctly and not interchanged
    anywhere in the script
````//

If any check fails: stop, report which rule ID
failed, do not emit geometry.
```

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### ## 8. CHANGE POLICY

If a new duty class, material, or standard revision is needed, add a new versioned block to Section 3 (e.g. `# v2 – heavy duty`) rather than editing values in place. Keep prior versions for traceability back to the standard/date they were sourced from.